

From Byproduct to Strategic Material: A Public-Markets Map

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Scope: Publicly traded companies operating across the four federally-backed recovery pathways, plus the Nevada facilities where several of these pathways concentrate

Federal money already flows toward the companies turning wastewater, ash, and biomass into aviation fuel, concrete, fertilizer, and filtration media. The Department of Energy, the Department of War (renamed from Defense in 2025), the EPA, the USDA, and the Department of Transportation each fund a piece of this economy for their own strategic reasons — supply-chain security, decarbonized infrastructure, or public-health protection. Below sits a working map of who trades publicly in each lane, checked against filings and reporting from this week rather than assumed from memory, since this sector moves fast: bankruptcies, spin-offs, and acquisitions have reshaped it substantially over the past year alone.

A note before the list. Several companies people commonly associate with this space — Redwood Materials, Fulcrum BioEnergy, Ostara, LanzaJet — remain privately held. Their public-market exposure runs through the larger companies partnering with or supplying them, which the entries below point toward. A second note: many of the pure-play names here are small, cash-burning, and volatile, true to an industry still building its first commercial-scale plants. This is a landscape map, rather than a recommendation, and none of it substitutes for a conversation with a financial professional who knows your full situation.

Part One — Four National Pathways

Carbon dioxide into sustainable aviation fuel

The Department of War and the Department of Energy back this pathway to secure fuel that never has to cross an ocean, while airlines chase the same fuel to meet their own

climate targets.

- **LanzaTech Global (NASDAQ: LNZA)** ferments waste carbon – industrial off-gas, municipal solid waste syngas – into ethanol at six commercial plants already running, then routes that ethanol toward jet fuel through its LanzaJet affiliate, which stays privately held even as LanzaTech grows its equity stake there.
- **Gevo (NASDAQ: GEVO)** is pursuing a DOE loan guarantee for its planned SAF facility and already holds a USDA Climate-Smart Farm to Flight grant, alongside a smaller, growing racing-fuel and demonstration-scale SAF business.
- **Aemetis (NASDAQ: AMTX)** is building its Riverbank, California SAF and renewable diesel plant on a former US Army ammunition site under the federal Base Realignment and Closure program, backed by roughly \$7 billion in signed airline offtake agreements.
- **Darling Ingredients (NYSE: DAR)** and **Valero Energy (NYSE: VLO)** co-own Diamond Green Diesel, whose Port Arthur and Norco plants already deliver blended SAF to JFK and other airports, built in part on Honeywell UOP's licensed Ecofining technology.
- **Honeywell (NASDAQ: HON)** and **Chevron (NYSE: CVX)** – the latter through its 2022 acquisition of Renewable Energy Group – round out the pathway as the technology licensor and the integrated refiner most exposed to the shift.

Mineral ash into geopolymers concrete and highway aggregates

Infrastructure law and Department of Transportation specifications increasingly favor low-carbon supplementary cementitious materials (SCMs) over straight Portland cement, and the company that once dominated this niche has just changed hands.

- **CRH plc (NYSE: CRH)** absorbed Eco Material Technologies, the largest fly-ash and SCM distributor in North America, in a 2025 acquisition. The SCM story now lives inside CRH's much larger public balance sheet rather than as its own ticker.
- **Amrize (NYSE/SIX: AMRZ)** carries Holcim's entire North American cement, aggregates, and building-envelope business as an independently traded company since its June 2025 spin-off, positioned to bid on federally funded highway and data-center work.
- **Vulcan Materials (NYSE: VMC)** and **Martin Marietta Materials (NYSE: MLM)**, the two largest US aggregates producers, both cite the Infrastructure Investment and Jobs Act's multi-year highway funding as a durable demand floor.

- **CEMEX (NYSE: CX), Heidelberg Materials (Frankfurt: HEI / OTC: HDELY), Eagle Materials (NYSE: EXP), and Arcosa (NYSE: ACA)** round out the field of major US-serving cement and aggregate producers.
- Worth naming and setting aside: **Summit Materials**, long a public aggregates and cement name, went private under Quikrete Holdings in February 2025 and no longer trades.

Nutrient precipitates into struvite

Rock phosphate is a finite, geopolitically concentrated resource, so USDA and DOE interest in domestic phosphorus recovery runs deep — yet the leading struvite technology itself, Ostara's Pearl system, stays privately held. Public exposure here runs through the water-technology companies that engineer and supply it, and through the conventional fertilizer majors the recovery movement aims to supplement.

- **Xylem (NYSE: XYL)** engineers and supplies Ostara's Pearl struvite reactors under a licensing partnership, and inherited a deep bench of nutrient-recovery and PFAS technology when it acquired Evoqua Water Technologies in 2023.
- **Veolia Environnement (Paris: VIE / OTC: VEOEY)** runs its own nutrient-recovery and water-technology lines at global scale, serving many of the same municipal wastewater clients.
- **Nutrien (NYSE/TSX: NTR), The Mosaic Company (NYSE: MOS), and CF Industries (NYSE: CF)** sit on the conventional side of the same policy conversation — the phosphate and nitrogen fertilizer majors whose supply chains domestic struvite recovery aims to diversify away from, rather than struvite producers themselves.

Sludge and digestate into biochar and PFAS remediation

The EPA's tightening PFAS drinking-water standard has turned activated carbon and biochar from a niche filtration product into critical infrastructure, and the smallest company on this list is also the one doing the most direct EPA-method biochar PFAS testing.

- **CHAR Technologies (TSXV: YES / OTC: CTRNF)** runs high-temperature pyrolysis in Ontario, has posted EPA-method test results showing no detectable PFAS in its biochar from biosolid feedstocks, and is piloting commercial-scale work with Baltimore's Department of Public Works.

- **Cabot Corporation (NYSE: CBT)** supplies granular activated carbon rated for PFOA and PFOS removal in public water systems.
 - **Kuraray Co. (TYO: 3405)** owns Calgon Carbon, the world's largest activated-carbon manufacturer, and is investing roughly \$100 million to expand PFAS-focused reactivated-carbon capacity at its Columbus, Ohio plant alone.
 - **Pentair (NYSE: PNR)** and **Veralto (NYSE: VLTO)** supply the granular activated carbon systems and the PFAS analytics, respectively, that utilities lean on to prove compliance.
 - Two much smaller, thinly traded names worth knowing rather than betting heavily on: **Verde Resources** and **Millennium Sustainable Ventures**, both early-stage biochar and activated-carbon developers trading as US microcaps.
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Part Two — Nevada, Where the Pathways Concentrate

Nevada earns its place in this conversation honestly: cheap land, water scarcity that forces closed-loop discipline, and a state government actively courting this exact category of industry.

Closed-loop battery metallurgy

- **American Battery Technology Company (NASDAQ: ABAT)**, headquartered in Reno, just had a \$115 million DOE grant reinstated on appeal for its Tonopah Flats lithium refinery, and carries a FAST-41 federal priority-project designation alongside partnerships with BASF, GM, Ford, and Stellantis through the US Advanced Battery Consortium.
- **Aqua Metals (NASDAQ: AQMS)**, also Reno-based and sited at the Tahoe-Reno Industrial Center itself, runs its AquaRefining electrochemical process there and was recently selected as the industrial partner on a DOE-funded Idaho National Laboratory research program.
- Worth naming clearly: **Redwood Materials**, TRIC's largest and best-known battery recycler, remains privately held with no near-term IPO signaled. **Li-Cycle**, once the sector's most visible public name, filed for bankruptcy in 2025; its shares now trade for fractions of a cent, and Glencore acquired its key assets through a credit bid.

Direct lithium extraction

- **Lithium Americas (NYSE/TSX: LAC)** owns 62% of the Thacker Pass joint venture with General Motors, backed by a \$2.26 billion DOE loan under which the department now also holds equity warrants in both the company and the project itself.
- **ioneer Ltd (NASDAQ: IONR)** secured a \$996 million DOE loan for its Rhyolite Ridge lithium-boron project and cleared a major legal challenge to its federal permits in March 2026.
- **Century Lithium (TSXV: LCE / OTCQX: CYDVF)** runs its direct lithium extraction pilot at Clayton Valley, next door to Albemarle's brine operation, with a feasibility study showing a projected \$4 billion net present value.
- **Albemarle (NYSE: ALB)** has drawn lithium from Silver Peak by solar evaporation for more than fifty years, the longest-running operation of its kind in the country.

Water purification and aquifer storage recovery

The Southern Nevada Water Authority itself is a public agency rather than a stock, yet the technology and engineering companies that build and run systems like it trade openly.

- **Xylem (NYSE: XYL), American Water Works (NYSE: AWK), Veolia (VEOEY), and Pentair (NYSE: PNR)** supply the membranes, pumps, and treatment trains behind advanced water reuse.
- **AECOM (NYSE: ACM) and Jacobs Solutions (NYSE: J)** design and operate large-scale potable reuse programs across the arid Southwest, the same discipline Nevada's own reclamation systems depend on.

Waste-to-fuels and municipal organic pyrolysis

Here the record calls for an honest correction rather than a stock pick. Fulcrum BioEnergy, the company behind the Sierra BioFuels plant that made this pathway famous, was always privately held, and it no longer operates: the company filed for Chapter 11 bankruptcy in September 2024, after shutting the Reno plant that June. The data-center developer Switch — itself currently private under DigitalBridge Group (NYSE: DBRG), though newly and confidentially filed for its own IPO — acquired the biorefinery out of bankruptcy, while a Waste Management affiliate bought the feedstock facility.

Waste Management (NYSE: WM) and the broader SAF names from Part One — LanzaTech, Gevo, and the Darling/Valero joint venture — carry the pathway's remaining public exposure.

Reading This List Honestly

Company status in this sector changes fast. Three of the names people most associate with these pathways — Redwood Materials, Fulcrum BioEnergy, Ostara — never traded publicly, or have since stopped. Two more, Li-Cycle and Summit Materials, traded publicly within the past two years and no longer do, for opposite reasons: one collapsed, one got bought at a premium. Treat every ticker above as a starting point for your own filing review rather than a settled fact, and treat the small-cap, pre-revenue names especially carefully — several run on DOE loans, grant reinstatements, and offtake agreements rather than current profit.

May this map serve the questions you bring to it, and may every number on it earn a second look before it earns a decision.